

SIMATS ENGINEERING
ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)

Total Marks: 25

Question Count: 5

Code of Conduct

I B Sandhya(192311292) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Sandhya

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.
Provide steps to ensure business continuity during failures.
4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data.
Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.
5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification

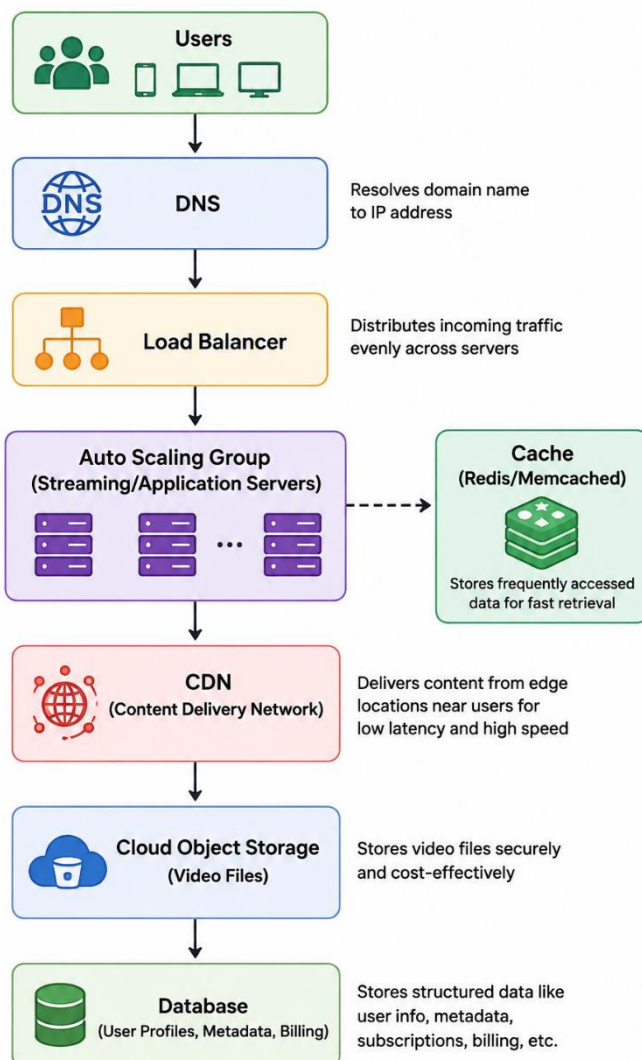
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented
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ANSWERS

1. Cloud Architecture for a Video Streaming Platform

1. Cloud Architecture Design

A scalable cloud architecture for a media company's video streaming platform should include the following components:



Components:

- **Load Balancer:** Distributes user requests evenly across multiple streaming servers.

- **Auto Scaling Group:** Automatically adds or removes servers based on demand.
- **CDN (Content Delivery Network):** Stores video content at edge locations close to users, reducing latency.
- **Cache (Redis/Memcached):** Stores frequently accessed video metadata, user sessions, and popular content.
- **Cloud Object Storage:** Stores video files reliably and cost-effectively.
- **Database:** Maintains user information, subscriptions, watch history, and video metadata.

2. How Load Balancing and Auto-Scaling Ensure Smooth Streaming

Load Balancing

- Distributes incoming traffic evenly among available servers.
- Prevents any single server from becoming overloaded.
- Improves availability and fault tolerance.
- Redirects traffic if a server fails.

Benefits:

- Reduced buffering
- Faster response times
- High availability
- Better user experience

Auto-Scaling

- Monitors CPU, memory, network traffic, or concurrent users.
- Automatically launches new instances during peak demand.
- Terminates unnecessary instances during low traffic.

Example:

- During a live cricket match, millions of users join.
- Auto-scaling increases streaming servers automatically.
- After the match, extra servers are removed to reduce costs.

Benefits:

- Handles sudden traffic spikes
- Maintains consistent streaming quality
- Eliminates manual server management
- Optimizes resource utilization

3. Cost Optimization Strategies

To reduce operational costs while maintaining performance:

1. **Use a CDN**
 - Delivers videos from edge locations.
 - Reduces bandwidth usage and server load.
2. **Auto-Scaling**

- Pay only for resources used.
- Avoid running idle servers.
- 3. **Object Storage**
 - Store videos in low-cost cloud storage.
 - Use archival storage for rarely viewed content.
- 4. **Caching**
 - Cache popular videos and metadata.
 - Reduces repeated database and storage access.
- 5. **Reserved or Savings Plans**
 - Use reserved instances for predictable workloads.
 - Use spot/preemptible instances for non-critical background tasks like video encoding.
- 6. **Video Compression**
 - Use efficient codecs (e.g., H.265/AV1) to reduce storage and bandwidth costs.
- 7. **Monitoring and Analytics**
 - Continuously monitor resource usage.
 - Remove unused resources and optimize capacity planning.

Conclusion

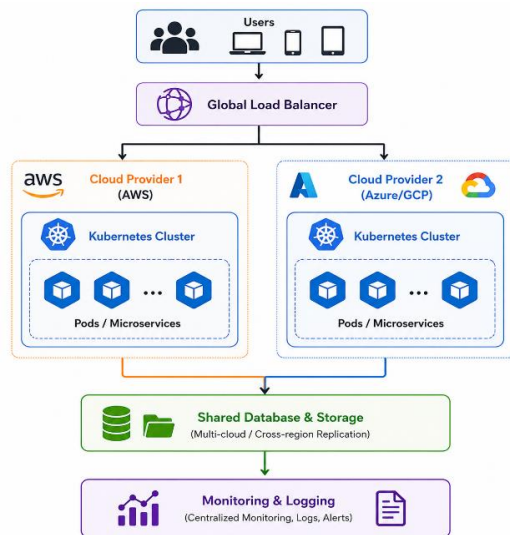
A cloud-based video streaming platform should combine **load balancing, auto-scaling, CDNs, caching, and cloud object storage** to efficiently serve millions of concurrent users. Load balancing distributes traffic evenly, auto-scaling adjusts resources automatically based on demand, caching and CDNs minimize latency and buffering, while cost optimization techniques such as object storage, compression, and reserved pricing help reduce operational expenses. This architecture provides **high availability, scalability, low latency, and cost efficiency**, ensuring a smooth streaming experience even during peak traffic.

2. FinTech Startup: Multi-Cloud Container Deployment

Solution Using Container Orchestration and Multi-Cloud

A fintech startup can use **Kubernetes (K8s)** as the container orchestration platform to deploy and manage applications consistently across multiple cloud providers (such as AWS, Azure, and Google Cloud).

Architecture



Service Discovery and Scaling

Service Discovery

- Kubernetes provides **built-in Service Discovery** using **Kubernetes Services** and **CoreDNS**.
- Each microservice gets a unique DNS name.
- Applications communicate using service names instead of IP addresses.
- This ensures reliable communication even when containers restart or move.

Scaling

- **Horizontal Pod Autoscaler (HPA)** automatically increases or decreases the number of pods based on CPU, memory, or custom metrics.
- **Cluster Autoscaler** adds or removes worker nodes when required.
- Load balancers distribute incoming traffic evenly across available pods.

Benefits

- Automatic scaling during peak demand.
- High availability.
- Better performance.
- Efficient resource utilization.

Benefits of Multi-Cloud Deployment

- **High Availability**
 - If one cloud provider experiences downtime, applications continue running on another provider.
- **Disaster Recovery**
 - Data and applications remain available even during regional failures.
- **Avoid Vendor Lock-in**
 - Applications are portable across different cloud platforms.
- **Improved Performance**

- Users connect to the nearest cloud region, reducing latency.
- **Cost Optimization**
 - Choose the most cost-effective cloud provider for different workloads.
- **Regulatory Compliance**
 - Store data in specific geographic regions to meet legal requirements.

Risks of Multi-Cloud Deployment

1. **Increased Management Complexity**
 - Managing multiple cloud platforms requires skilled administrators.
2. **Higher Operational Costs**
 - Networking, monitoring, and management costs may increase.
3. **Security Challenges**
 - Different cloud providers have different security policies and configurations.
4. **Data Synchronization Issues**
 - Keeping databases consistent across clouds can be difficult.
5. **Network Latency**
 - Communication between cloud providers may introduce delays.
6. **Monitoring Complexity**
 - Requires centralized monitoring and logging tools to manage all environments.

Conclusion

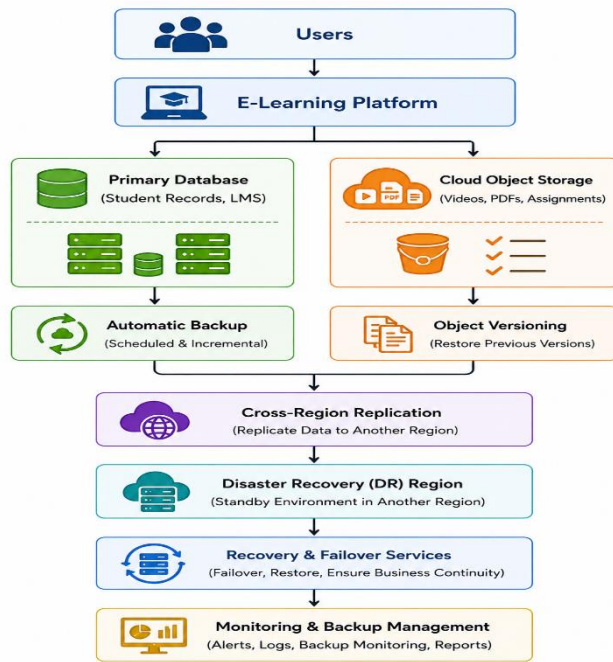
Using **Kubernetes** with a **multi-cloud architecture** enables consistent application deployment, automated scaling, and efficient service discovery across multiple cloud providers. Service discovery through Kubernetes Services and CoreDNS ensures seamless communication, while autoscaling maintains application performance during varying workloads. Although multi-cloud offers **high availability, disaster recovery, vendor independence, and cost optimization**, it also introduces challenges such as **greater operational complexity, security management, and cross-cloud networking**, which can be addressed with proper orchestration and centralized monitoring.

3. Cloud Backup and Disaster Recovery for an E-Learning Platform

Solution Using Cloud-Native Backup, Replication, and Versioning

An e-learning platform can use cloud-native services to protect against accidental data deletion and system failures.

Architecture



Recovery Time Objective (RTO) and Recovery Point Objective (RPO)

Recovery Time Objective (RTO)

- **Definition:** The maximum acceptable time to restore services after a failure.
- **How it is achieved:**
 - Automatic failover to the disaster recovery region.
 - Pre-configured standby servers.
 - Automated recovery scripts.
 - Continuous monitoring.

Example:

If the RTO is **15 minutes**, the e-learning platform should become operational within 15 minutes after an outage.

Recovery Point Objective (RPO)

- **Definition:** The maximum acceptable amount of data loss measured in time.
- **How it is achieved:**
 - Continuous data replication.
 - Frequent automatic backups.
 - Object versioning.
 - Database snapshots.

Example:

If the RPO is **5 minutes**, at most 5 minutes of data (such as quiz submissions or uploaded assignments) can be lost.

Business Continuity Steps

1. **Enable Automatic Backups**

- Schedule daily full backups and frequent incremental backups.
- 2. **Enable Object Versioning**
 - Recover accidentally deleted or modified files.
- 3. **Use Cross-Region Replication**
 - Replicate data to another geographic region for disaster recovery.
- 4. **Implement Disaster Recovery Site**
 - Maintain a standby environment ready for failover.
- 5. **Configure Automatic Failover**
 - Redirect users to the DR environment if the primary site fails.
- 6. **Monitor System Health**
 - Use cloud monitoring and alerts to detect failures quickly.
- 7. **Regular Disaster Recovery Testing**
 - Perform backup restoration and failover drills periodically.
- 8. **Secure Backups**
 - Encrypt backup data and restrict access using Identity and Access Management (IAM).

Benefits

- Protects against accidental deletion.
- Ensures minimal data loss.
- Reduces service downtime.
- Provides high availability.
- Supports quick disaster recovery.
- Improves reliability and user trust.

Conclusion

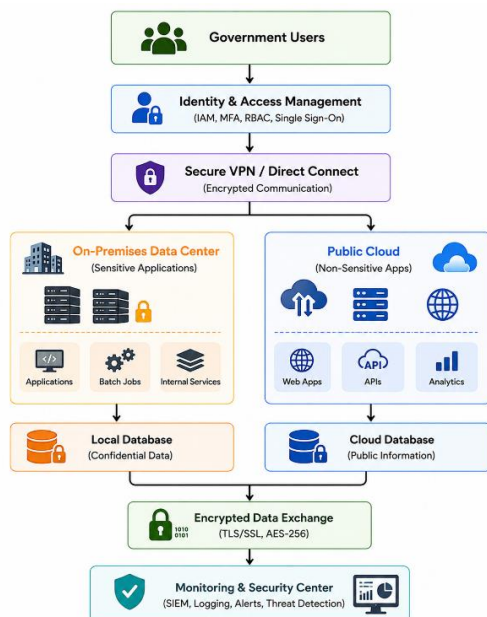
A robust disaster recovery strategy for an e-learning platform combines **cloud-native backups, object versioning, and cross-region replication** to safeguard critical data. Continuous replication and frequent backups help achieve a low **RPO**, while automated failover and standby resources ensure a low **RTO**. By implementing regular backup testing, monitoring, encryption, and disaster recovery planning, the organization can maintain **business continuity**, minimize downtime, and ensure uninterrupted learning services even during failures.

4. Hybrid Cloud Architecture for a Government Agency

Hybrid Cloud Architecture Design

A government agency can adopt a **Hybrid Cloud** model where **sensitive applications and data remain on-premises** to meet compliance requirements, while **non-sensitive applications** are hosted in the public cloud for scalability and cost efficiency.

Architecture



Secure Communication Between Environments

To ensure secure communication between the on-premises data center and the public cloud:

- Use **VPN (Virtual Private Network)** or **Dedicated Direct Connect/ExpressRoute** for secure connectivity.
- Encrypt all data in transit using **TLS/SSL** protocols.
- Deploy **Firewalls** and **Network Security Groups (NSGs)** to control network traffic.
- Use **API Gateways** for secure communication between applications.
- Continuously monitor network activity using cloud security monitoring tools.

Identity Management and Encryption

Identity Management

- Implement **Identity and Access Management (IAM)** for centralized authentication.
- Enable **Multi-Factor Authentication (MFA)** for all users.
- Use **Role-Based Access Control (RBAC)** to provide least-privilege access.
- Integrate on-premises directories (e.g., Active Directory) with cloud identity services.
- Maintain audit logs for user activities.

Encryption

Data at Rest

- Encrypt databases, storage, and backups using **AES-256** encryption.

Data in Transit

- Protect communication using **TLS/SSL** encryption.

Key Management

- Store and manage encryption keys using a **Key Management Service (KMS)** or Hardware Security Module (HSM).
- Rotate encryption keys regularly.

Challenges in Managing Hybrid Cloud Environments

- **Security Complexity**
 - Managing consistent security policies across on-premises and cloud environments.
- **Compliance Requirements**
 - Ensuring sensitive government data complies with legal and regulatory standards.
- **Network Latency**
 - Communication between on-premises systems and the cloud may introduce delays.
- **Identity Synchronization**
 - Keeping user identities and permissions consistent across environments.
- **Data Integration**
 - Synchronizing data between on-premises databases and cloud storage.
- **Monitoring and Management**
 - Requires centralized monitoring tools for visibility across both environments.
- **Disaster Recovery**
 - Coordinating backup and recovery processes across hybrid infrastructure.

Benefits of Hybrid Cloud

- Protects sensitive government data.
- Meets regulatory and compliance requirements.
- Improves scalability for public-facing applications.
- Reduces infrastructure costs.
- Enables flexible resource allocation.

- Provides business continuity and disaster recovery.

Conclusion

A **Hybrid Cloud** architecture enables government agencies to securely manage **sensitive data on-premises** while leveraging the **public cloud** for non-sensitive workloads. Secure communication is achieved using VPNs or dedicated private connections with TLS encryption, while centralized IAM, MFA, RBAC, and encryption protect users and data. Although hybrid cloud environments introduce challenges such as security management, compliance, and data synchronization, they provide an effective balance of **security, scalability, compliance, and cost efficiency**.

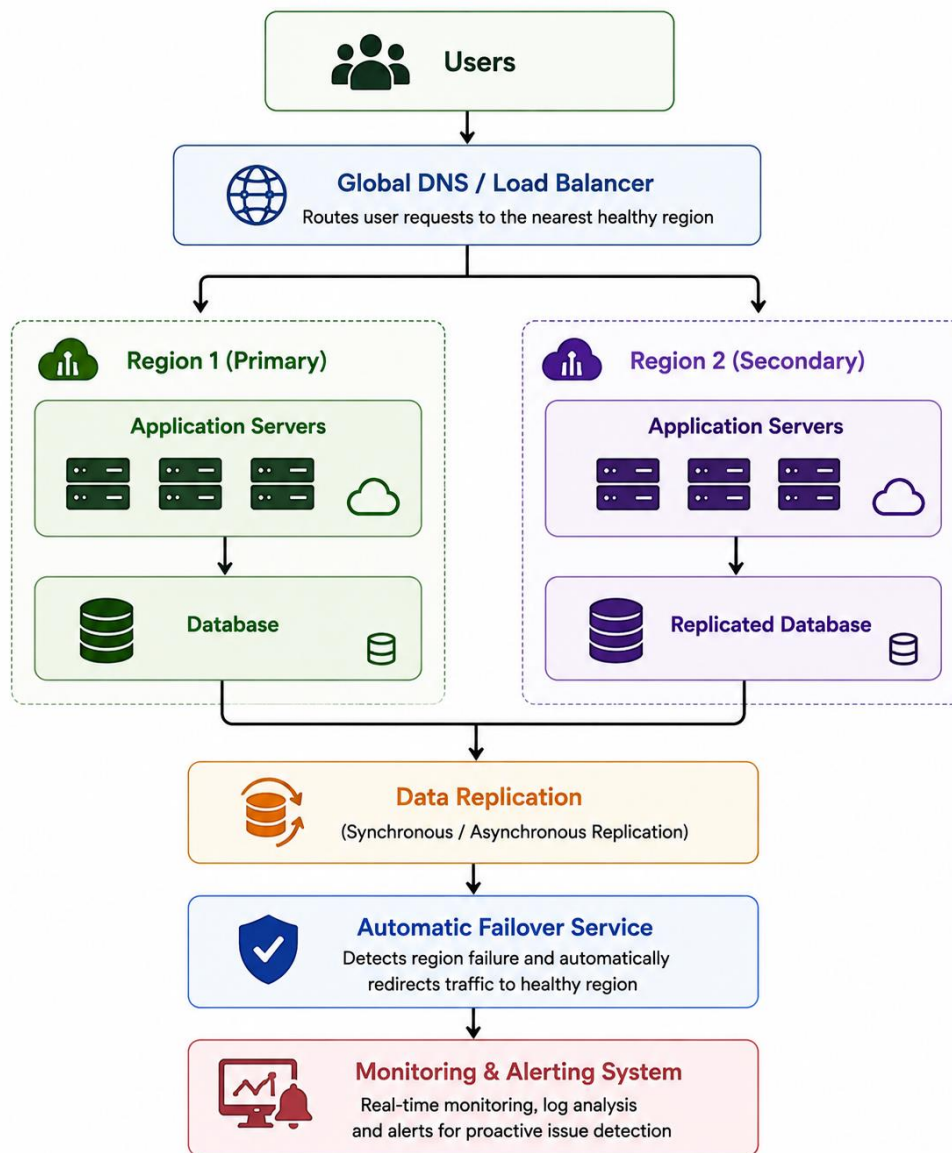
6. Multi-Region Deployment for a SaaS Provider

Multi-Region Deployment Strategy

To ensure **high availability** and **fault tolerance**, the SaaS provider should deploy its application across **multiple cloud regions** using an **Active-Active** or **Active-Passive** architecture.

- **Primary Region:** Handles user traffic under normal conditions.
- **Secondary Region:** Acts as a backup (Active-Passive) or shares traffic (Active-Active).
- **Global Load Balancer/DNS:** Routes users to the nearest healthy region.
- **Database Replication:** Keeps data synchronized between regions.
- **Monitoring & Alerting:** Continuously checks application and infrastructure health.

Architecture



Data Replication and Failover

Data Replication

- Use **synchronous replication** for critical data requiring zero or minimal data loss.
- Use **asynchronous replication** for geographically distant regions to reduce latency.
- Enable **continuous database replication** and regular backups.
- Store application files in **cross-region replicated object storage**.

Benefits

- Prevents data loss.
- Keeps both regions synchronized.
- Supports quick recovery after failures.

Failover Mechanism

- Configure **health checks** to monitor application availability.
- If the primary region becomes unavailable:
 - Global DNS or Load Balancer automatically redirects traffic to the secondary region.
- Use **automatic failover** for databases and application servers.
- Restore the primary region once it becomes healthy.

Benefits

- Minimal downtime.
- Seamless user experience.
- Automatic recovery without manual intervention.

Monitoring and Alerting Strategies

Monitoring

- Monitor:
 - CPU and Memory usage
 - Network latency
 - Application response time
 - Database performance
 - Error rates
 - Disk utilization
 - Service availability

Alerting

- Configure alerts for:
 - High CPU or memory utilization
 - Application failures
 - Database replication delays
 - Region outages
 - Network connectivity issues
 - Security incidents

Best Practices

- Use centralized dashboards for all regions.
- Enable real-time log collection and analysis.
- Configure automated notifications via email, SMS, or collaboration tools.
- Perform regular health checks and disaster recovery drills.
- Use AI-based anomaly detection to identify unusual behavior early.

Benefits of Multi-Region Deployment

- High Availability
- Fault Tolerance
- Disaster Recovery
- Low Latency for Global Users
- Improved Reliability

- Business Continuity
- Better User Experience

Conclusion

A **multi-region cloud deployment** ensures that a SaaS application remains available even if an entire region fails. Global load balancing distributes traffic across regions, while continuous data replication keeps databases synchronized. Automatic failover mechanisms redirect users to healthy regions during outages, minimizing downtime. Combined with comprehensive monitoring, real-time alerting, and regular disaster recovery testing, this architecture provides **high availability, fault tolerance, business continuity, and a seamless user experience**.